

Ideation Phase

Define the Problem Statements

Date	1 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Customer Problem Statement

Project: Rising Waters – Intelligent Flood Prediction and Early Warning System

Floods often occur with little warning, causing loss of life, property damage, and disruption to communities. Disaster management authorities and local communities need a reliable way to predict flood risks before they happen. Current prediction methods may not always provide timely or accurate information, making it difficult to plan evacuations and allocate emergency resources effectively. Our solution, **Rising Waters**, is a Machine Learning-based flood prediction system that uses historical weather and rainfall data to predict flood risk. The system provides early warnings through a simple web application, helping authorities make faster decisions and improve disaster preparedness. A clear understanding of this customer problem helps our team develop a practical, accurate, and user-friendly solution that meets the needs of disaster management organizations and communities at risk of flooding.

Customer Problem Statement – Rising Waters: Intelligent Flood Prediction and Early Warning System

 I am	Disaster Management Officer / Meteorologist responsible for monitoring weather and protecting communities from flood risks.	
 I'm trying to	Predict flood occurrences early using weather and rainfall data to plan evacuations, allocate resources, and issue timely warnings.	
 but	Current prediction methods are not always accurate or timely, making it difficult to identify high-risk areas before flooding occurs.	
 because	Weather conditions change rapidly, historical data is complex, and traditional forecasting methods cannot effectively analyze multiple influencing factors.	
 which makes me feel	Concerned about public safety and the ability to make fast, informed decisions during flood emergencies.	

Example:

I am a Disaster Management Officer	I'm trying to predict flood risk early and protect communities	but weather data is difficult to analyze quickly.	because traditional methods are slow and lack accurate predictions.	which makes me feel Concerned
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Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
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PS-1	A Disaster Management Officer	Predict flood-prone areas and issue early warnings	I do not receive accurate and timely flood predictions	Existing systems rely on limited data and manual monitoring	Concerned about public safety and emergency response
PS-2	A Meteorologists	Analyze weather and rainfall data to forecast floods	It is difficult to process large amounts of historical weather data manually	Traditional analysis is time-consuming and less efficient	Frustrated and unable to provide quick, reliable forecasts